
BARBIE User Manual

DESCRIPTION

The BARBIE (Buoy for Antarctic Research and Basic Information Exchange) is a device intended to address some common concerns with regards to research in the Antarctic. In particular, it attempts to address:

- Power supply
- Data storage
- Data transmission / retrieval
- Data security
- Water proofing
- Protection against collisions

The project's documentation, PCB design and code can be found on GitHub:

<https://github.com/Adedamola-Yusuff/EEE4113F-Project-Group-12>

TYPICAL APPLICATION

The BARBIE interfaces internally with a custom sensor module (designed by the user). This sensor module receives power from the BARBIE, and sends data over the I2C interface. The BARBIE automatically stores this data in the micro SD card, and eventually distributes the data to other nearby BARBIEs. The BARBIE will also transmit the data to the TEA software over satellite, along with its GPS location. Using the TEA software nearby, a data-dump can be requested, in which the BARBIE dumps all of its data over a Wi-Fi connection to the user's device.

NOTES

The BARBIE is still experimental, and not fully tested.

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DIMENSIONS

Dimension	Value	Unit
Height	500	mm
Diameter	272	mm

BATTERY SPECIFICATIONS

Characteristic	Value	Unit
Minimum Voltage	6	V
Maximum Voltage	15	V
Recommended Chemistry	Lithium Titanate Oxide	-

NETWORKING PERFORMANCE

Protocol	Metric	Value	Unit
Wi-Fi	Maximum Range	10	m
Wi-Fi	Data Rate	250	kb/s
Global Positioning System	Tolerance	10	m

POWERING SENSOR MODULE

Supply	Rating	Value	Unit
Primary	Voltage	3.3	V
Primary	Current	300	mA
Secondary	Voltage	5	V
Secondary	Current	300	mA

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COLLECTING DATA

The onboard ESP32 is in light sleep most of the time. The sensor module is required to raise the “data ready” pin to wake up the ESP32. Give 5 milliseconds.

After being woken up, the ESP32 gives the sensor module 500 milliseconds to send the header of the I2C Data Format (the write address and the payload length). The ESP32 then gives another 500 milliseconds to transmit the payload. Failure to meet either of these deadlines will result in the ESP32 discarding what it received and going back to sleep.

The write address 0x01 corresponds to giving data.

The write address 0x02 corresponds to giving instructions (not yet implemented).

DATA FORMATS

Data packet

0x00	BARBIE ID
0x01	Sequence Number
0x02 - 0xFD	Data from Sensor Module
0xFE - 0xFF	16-bit Cyclic Redundancy Check / IBM Convention

I2C Data Format

0x00	Write Address
0x01	Payload Length
0x02 - ...	Payload

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CONFIGURABLE PROPERTIES

Several behaviours/properties of the BARBIE can be configured by flashing the onboard ESP32 with a modified `software_config.h` file.

Property	Macro Name	Default Value	Notes
Address of BARBIE when acting as an I2C slave	I2C_SLAVE_ADDRESS	0x67	-
Sharing data with peers	DATA_SHARING	N/A (enabled)	Comment out this macro to stop the BARBIE from sharing data to peers
Period of sharing data with peers	SHARE_PERIOD	1ULL*60*60*1000*1000	To use this macro to modify the period, the macro <code>SHARE_PERIOD_OVERRIDE</code> must be uncommented
Sending GPS location	SEND_GPS	N/A (enabled)	Comment out this macro to stop the BARBIE from sending its GPS
Transmitting data over satellite	TRANSMIT_DATA	N/A (enabled)	Comment out this macro to stop the BARBIE from sending data over satellite
Period of transmitting data and GPS over satellite	SATELLITE_PERIOD	24ULL*60*60*1000*1000	-
The identification number of the BARBIE	DEVICE_ID	0	To use this macro to modify the ID, the macro <code>DEVICE_ID_OVERRIDE</code> must be uncommented
Filepath to where data is stored on micro SD card	DATA_FILENAME_PATH	"/sdcard/data.bin"	-